Shu Yang

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Research Interests & Personal Background

Shu Yang is an undergraduate student in the Excellence Class of Computer Science at Nankai University. During high school, he won a bronze medal at the National Olympiad in Informatics (NOI 2022). After entering university, he continued to participate in ACM-ICPC/CCPC programming contests, winning multiple medals, and possesses solid skills in algorithm design and implementation. His long-term experience in programming contests has sparked a strong interest in the theoretical foundations of computational complexity and the intrinsic laws of algorithm design. He aspires to pursue foundational research in algorithm design in the future.

During his undergraduate studies, he has systematically taken theoretical courses such as *Computational Theory*, *Lattice-based Post-Quantum Cryptography*, *Algorithm Design and Analysis*, and *Randomized Algorithms*, as well as engineering practice courses like *Operating Systems*, *Compilers*, *Database Systems*, and *Computer Graphics*. He is capable of analyzing and solving complex problems, and also possesses engineering coding ability. Additionally, as an undergraduate research assistant, under the supervision of Associate Professor Shenwei Huang and Dr. Yidong Zhou, he participated in research on the solvability of graph coloring problems, gaining experience in literature reading, algorithmic reduction, and complexity analysis.

Education

Nankai University, College of Computer Science, Computer Science and Technology (Excellence Class) *B.S. (expected)* 2023.09 – Present

- GPA: 3.48/4.0, Rank: Top 38%

Affiliated High School of Shanxi University, Experimental Class 2020.09 – 2023.06

- Participated in informatics competitions during high school, ranked third in the province and selected for the Shanxi provincial team, won a bronze medal at NOI, laying the foundation for subsequent university studies.

Research Experience

Complexity of 3-Coloring on a Special Graph Class (Supervisors: Associate Professor Shenwei Huang, Dr. Yidong Zhou)

- **Goal**: Determine whether the 3-coloring problem on a class of graphs with specific structural constraints is in P.
- **Main Work**:
  - Reviewed and summarized core papers on the complexity boundary of graph coloring from the past decade, wrote literature review notes.
  - Attempted to design polynomial-time algorithms: used graph decomposition techniques to reduce the target graph class, initially proposed a 3-coloring algorithm framework.
  - Explored reduction proofs: constructed polynomial reductions from known NP-complete problems (e.g., 3-SAT or Not-All-Equal 3-SAT) to this graph class.
- **Academic Communication**: Attended weekly group seminars, reported research progress, and discussed verification methods for correctness of reductions with supervisors.

Course Projects

Implementation of a Mini Compiler for SysY

- Implemented a front-end compiler for the SysY language (a C subset) using C++ and Flex/Bison: lexical analysis, syntax analysis, semantic analysis, code optimization, and intermediate code generation.
- Designed symbol tables and type checking modules, implemented intermediate representation for basic control flow and function calls.

Database System Based on the CMU BUSTUB Framework

- Implemented in-memory/disk dual storage model using C++, supporting B+ tree indexes, a SQL subset (SELECT, INSERT, DELETE), and ACID transaction properties (simple log recovery).
- Buffer LRU replacement policy; performance comparison test reports for single-table scans showed an approximately 40x speedup for indexed queries.

Real-time Ray Tracing Renderer

- Implemented ray tracing in C++, supporting multiple light sources, shadows, specular reflection, and refraction.
- Used KD-Tree to accelerate intersection tests, greatly improving rendering speed.

All projects above are uploaded to my GitHub repository.

Competition Experience

In the ACM three-person team, I am mainly responsible for dynamic programming, graph theory, combinatorial counting, and some data structure design. I excel at problems requiring algorithmic thinking.

ACM-ICPC International Collegiate Programming Contest 2025

- Shenyang Regional: Silver medal (team rank 2).
- Hong Kong Regional: Silver medal (team rank 3).
- EC Final: Bronze medal.

CCPC China Collegiate Programming Contest 2025

- Zhengzhou Site: Silver medal; CCPC Final: Bronze medal.

National Olympiad in Informatics (NOI 2022) 2022

- Bronze medal.

Technical Skills

- **Programming Languages**: Proficient in C/C++, Python.
- **Tools & Environments**: Linux (Ubuntu), Git, VS Code, LaTeX.
- **Theoretical Tools**: Algorithm complexity analysis, reduction proofs, combinatorics, basics of probabilistic methods.

English Proficiency

IELTS Overall Score 7.0 April 2025

- Listening 7.5, Reading 8, Writing 6.5, Speaking 6.5.
- Capable of reading English academic papers fluently and writing technical documents.

Honors and Awards

- Bronze Medal, National Olympiad in Informatics (NOI 2022)
- Multiple Medals, International Collegiate Programming Contest (ICPC) / China Collegiate Programming Contest (CCPC)

Undergraduate Academic Transcript

Name: Yang Shu	Student ID: 2311673	Admission Date: 2023.09	Length of Schooling: 4 Years
Department: College of Computer Science		Major: Computer Science and Technology	

Course Title	Grade	Credit	Grade point
Fall 2023			
Linear Algebra	C+	4.0	2.3
A Seminar for New NKUers	P	1.0	/
Database Technology and Application	A	2.0	4.0
Higher Mathematics Class AI	B-	5.0	2.7
Football Beginner's Class	A-	1.0	3.7
College Students' Mental Health	P	2.0	/
High Level Language Program Design2-1	A	3.5	4.0
Graph Algorithms in life	P	1.0	/
Ideology and morality and rule of law	C+	2.5	2.3
College English I A	A	2.0	4.0
Spring 2024			
Higher Mathematics Class AII	B	5.0	3.0
Basic Principle of Marxism	C	2.5	2.0
College Physics (1)	A	4.0	4.0
Basic Physics Experiment	B+	2.0	3.3
Outdoor Activity	P	1.0	/
Food, Culture and Health	A	2.0	4.0
Chorus and Conducting	A-	1.0	3.7
Military Theory	A-	2.0	3.7
Mathematical Culture	A-	2.0	3.7
Swimming Advanced Class	A	1.0	4.0
College English II A	A-	2.0	3.7
Probability and Mathematical Statistics	A-	4.0	3.7
High Level Language Program Design2-2	A-	2.5	3.7
Fall 2024			
Data Structure	A	3.5	4.0
Assembly Language and Reverse Engineering	C	2.5	2.0
Verilog Programming Design	A-	1.0	3.7
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Course Title	Grade	Credit	Grade point
Python for Artificial Intelligence	C+	2.5	2.3
Art and Aesthetic	P	2.0	/
Military skill training	A	2.0	4.0
Digital Logic	A-	3.5	3.7
Discrete mathematics	A-	4.0	3.7
College Chinese	B+	2.0	3.3
Digital Signal Processing	A-	2.0	3.7
Tennis Beginner's Class	A-	1.0	3.7
Essentials of the Modern History of China	A-	2.5	3.7
Spring 2025			
Introduction to Mao Zedong's Thoughts and the Theory of Socialism with Chinese Characteristics	A	2.5	4.0
Badminton Beginner's Class	A	1.0	4.0
Principles of Computer Organization	B+	3.5	3.3
Introduction to Algorithms	A	3.5	4.0
Database System	B+	3.5	3.3
Introduction to Artificial Intelligence	A-	2.5	3.7
Foundation of Cryptography	B	3.0	3.0
Introduction to Parallel Programming	A	2.5	4.0
Critical Thinking and English Communication	A	2.0	4.0
Software Security	B+	2.5	3.3
Fall 2025			
Game Theory	A	2.0	4.0
Introduction to Philosophy	P	2.0	/
Project Training and Practice	A	2.0	4.0
Computer Architecture	B+	3.5	3.3
Computer Graphics	B+	2.5	3.3
Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	B-	3.0	2.7
Computer Networks	B+	3.5	3.3
Principles of Compilers	A	3.5	4.0
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Official Seal:

Issue Date: 2026-06-01

南開大學 Nankai University

Undergraduate Academic Transcript

Name: Yang Shu		Student ID: 2311673		Admission Date: 2023.09		Length of Schooling: 4 Years	
Department: College of Computer Science				Major: Computer Science and Technology			
Course Title	Grade	Credit	Grade point	Course Title	Grade	Credit	Grade point
Operating System	A-	3.5	3.7				
Lattices in Computer Sciecnes	A	2.0	4.0				
Global outlook and personal development	A-	1.0	3.7				
---End of Transcript---							
Earned Credits: 138.5				GPA: 3.456/4			

Official Seal:

Issue Date: 2026-06-01

南开大学本科生成绩单说明

一、成绩设置

百分制成绩（X） （60 及以上为及格）	等级制成绩 （D 级以上为及格）	成绩绩点 （1.0 以上为及格）	两级制成绩 （通过为及格）
90≤X	A	4.0	通过
85≤X＜90	A-	3.7	
81≤X＜85	B+	3.3	
78≤X＜81	B	3.0	
75≤X＜78	B-	2.7	
72≤X＜75	C+	2.3	
69≤X＜72	C	2.0	
66≤X＜69	C-	1.7	
63≤X＜66	D+	1.3	
60≤X＜63	D	1.0	
X＜60	F	0	不通过

二、平均学分绩

课程学分绩=学分数×分数

平均学分绩=∑已修课程学分绩/∑已修课程学分

自 2023 级本科生起，学校不再提供平均学分绩证明。

三、平均学分绩点（GPA）

课程学分绩点=课程成绩等级对应绩点×课程学分

平均学分绩点（GPA）=∑已修课程学分绩点/∑已修课程学分

自 2017 级本科生起，学校实行成绩等级及绩点制。

四、符号注释

符号	注释
P	通过
F	未通过
W	退课、未完成
△	校外交流获得学分
*	全英文课程

注明:“通过”“未通过”“退课”“未完成”及“校外交流获得学分”的成绩均不参与学分绩、学分绩点计算。

Nankai University

Description of Undergraduate Transcript

1. Grade Scheme

Percentage Grades (X) (Passing grades: 60 and above)	Letter Grades (Passing grades: D and above)	Grade Points (Passing grades: 1.0 and above)	Pass/Fail Grading System
90≤X	A	4.0	Pass
85≤X＜90	A-	3.7	
81≤X＜85	B+	3.3	
78≤X＜81	B	3.0	
75≤X＜78	B-	2.7	
72≤X＜75	C+	2.3	
69≤X＜72	C	2.0	
66≤X＜69	C-	1.7	
63≤X＜66	D+	1.3	
60≤X＜63	D	1.0	
X＜60	F	0	Fail

2. Cumulative Weighted Average (CWA)

Course Credit Grade = Credits × Score

Cumulative Weighted Average = ∑Credit Grades of Completed Courses / ∑Credits of Completed Courses

Since the class of 2023, the university will no longer provide the proof of cumulative weighted average.

3. Grade Point Average (GPA)

Course Grade Point = Corresponding Grade Point for Course Grade × Course Credits

Grade Point Average (GPA) = ∑Grade Points of Completed Courses / ∑Credits of Completed Courses

Since the class of 2017, the university has implemented Letter Grades and Grade Points system for undergraduates.

4. Symbol

Symbol	Description
P	Pass
F	Fail
W	Withdrawn / Incomplete
△	Credit Transfer
*	English Taught Course

Notes: Courses marked with P(Pass), F(Fail), W(Withdrawn / Incomplete) and △(Credit Transfer) are not taken into account in the CWA and GPA calculations.